Boundary integral framework for G_substrate derivation via Shilov ∂_S D_IV^5 = $(S^4 \times S^1)/\mathbb{Z}_2$. Casey-Keeper conversation: gravity emerges from 2D→3D radial lift × Shilov boundary correction. Explicit factorization: S^4 provides spatial 4/3 disk-to-ball lift; S^1 provides temporal/phase correction; $\mathbb{Z}_2$ provides Z_2-even K-type projection. Hua-Korányi reproduction formula in Bergman/Poisson-Szegő structure. Elie Lane G-B refined target: compute (4/3) recovery from S^4 + S^1 correction explicitly + Z_2-even K-type sum.

Keeper (Sunday 2026-05-31 EOD post-team)

Boundary Integral Investigation Framework — G from ∂_S D_IV^5

0. The Casey-Keeper conversation

Casey's insight cascade Sunday EOD refined the G chain:

Insight 1: The 24% miss IS the 2D→3D geometric lift (factor 4/3 = disk-to-ball volume ratio coefficient).

Insight 2: Of course you integrate by radius to lift — the substantive math is at the BOUNDARY. The boundary geometry term is exactly the remainder.

Insight 3: ∂_S has TWO factors ($S^4 \times S^1/\mathbb{Z}_2$). The sum/difference/product of their contributions provides the boundary correction.

This document sets up the explicit boundary integration framework.

1. Shilov boundary structure

∂_S D_IV^5 = $(S^4 \times S^1)/\mathbb{Z}_2$

Three distinct geometric contributions:

Factor	Dim	Volume	Physical role
S^4	4 real	$8\pi^2/3$	Spatial / Newtonian-geometric
$S^1/\mathbb{Z}_2$	1 real	π	Temporal / phase / U(1)
$\mathbb{Z}_2$	(identification)	—	CP / parity constraint coupling $S^4 \otimes S^1$

$\mathbb{Z}_2$ identification: $(x, \theta) \sim (-x, \theta + \pi)$ — antipodal map on S^4 combined with half-rotation on S^1.

Total Vol(∂_S) = $(8\pi^2/3) \times \pi = 8\pi^3/3$ (before c_FK normalization)

FK-normalized total: c_FK × $8\pi^3/3$ = (225/π^{9/2}) × ($8\pi^3/3$) = 600/π^{3/2} ≈ 107.7

2. Bergman kernel on D_IV^5 (RECALLED — needs FK Ch. XII-XIII verified citation)

```
K_Bergman(z, w̄) = c_FK / (1 − 2 z·w̄ + (z·z)(w̄·w̄))^{n_C}
                = c_FK / (1 − 2 z·w̄ + (z·z)(w̄·w̄))^5
```

Where: - $z \cdot \bar{w} = z_1\bar{w}_1 + \ldots + z_5\bar{w}_5$ (Hermitian inner product) - $z \cdot z = z_1^2 + \ldots + z_5^2$ (complex bilinear, no conjugation) - c_FK = 225/π^{9/2} (Saturday RATIFIED T2442) - Exponent n_C = 5 matches κ_Bergman = -n_C (Sunday K204-PARTIAL)

3. Poisson-Szegő kernel (RECALLED)

For boundary-to-interior reproduction:

$$P(z, \zeta) = |K(z, \bar{\zeta})|^2 / K(z, \bar{z})$$

At z = 0 (origin of D_IV^5):

$$P(0, \zeta) = c\_FK^2 / c\_FK = c\_FK \quad \text{(constant on } \partial\_S)$$

This gives the boundary mean-value identity:

$$f(0) = c\_FK \cdot \int_{\partial\_S} f(\zeta)\, d\mu(\zeta)$$

(With appropriate normalization of dμ as a probability measure: f(0) = ∫ f dμ_prob.)

4. K-invariant measure on ∂_S

$$d\mu(\zeta) = (3/(16\pi^3)) \cdot d\theta \cdot d\sigma_{S^4}(x)$$

(Probability normalized: ∫ dμ = 1.)

In FK-Hua normalization (unnormalized):

$$d\mu\_Hua(\zeta) = c\_FK \cdot d\theta \cdot d\sigma_{S^4}(x) \cdot \text{(phase normalization)}$$

5. The gravitational lift formula

Structural form:

$$G\_observed = (S^4 \text{ geometric lift}) \times (S^1 \text{ phase correction}) \times (\mathbb{Z}_2\text{−even substrate source}) \times G\_2D\_substrate$$

Where: - G_2D_substrate = ℏc · α^{N_c·g}/m_e^2 (Lyra's substrate formula, now reinterpreted as 2D-substrate G) - S^4 geometric lift = 4/3 (Casey insight: disk-to-ball volume coefficient) - S^1 phase correction = (Bergman-weighted $S^1/\mathbb{Z}_2$ measure on relevant substrate source) ≈ small ~1-3% - $\mathbb{Z}_2$-even projection = which K-types couple to gravity (those with even parity under $(x,\theta) \to (-x, \theta+\pi)$)

6. Explicit computation targets (Elie Lane G-B refined)

6a. S^4 geometric lift recovery

Compute: ∫_{S^4} (volume form) dσ → recover the 4/3 disk-to-ball lift factor

Method: - Vol(S^4 unit) = $8\pi^2/3$ - Project to relevant 3D-physical structure - Ratio Vol(S^4)/Vol($S^2 \times 2\pi$) = $(8\pi^2/3)/(8\pi^2) = 1/3$ — wait, that's not 4/3 - Or: integrate radial measure: ∫_0^1 (S^2 of radius r) dr = ∫_0^1 $4\pi r^2$ dr = (4/3)π ▯ - The 4/3 emerges from radial integration of S^2 over radius — this is the disk-to-ball lift

Verification target: explicit formula showing 4/3 from S^4-projection or radial integration in Bergman setup.

6b. S^1 phase correction

Compute: ∫_{$S^1/\mathbb{Z}_2$} (Bergman-weighted phase kernel) dθ

Candidate forms:

Form A: pure S^1 measure correction

```
S¹_correction_A = π/(2π) = 1/2
```

(too large — would give 50% correction, doesn't match 1-3% residual)

Form B: Z_2-even phase mode integral

```
S¹_correction_B = ∫₀^π e^{2inθ} dθ / π = δ_{n,0}
```

(only zero-mode survives — too restrictive)

Form C: Bergman-weighted phase integral

```
S¹_correction_C = (1/π) · ∫₀^π K_Bergman_factor(e^{iθ}) dθ
```

(depends on Bergman kernel at boundary phases — concrete computation needed)

Form D: Heat-semigroup phase integral (Lyra Sunday-morning operator framework)

```
S¹_correction_D = (1/π) · ∫₀^π exp(−τ_S¹ · ⟨H_B⟩) dθ
```

(connects to Tier 0 commitment operator; multi-week)

Verification target: which form gives the actual numerical residual? Cross-check (4/3) × (S^1_correction) against G_observed/(ℏc · α^{N_c·g}/m_e^2) ratio.

6c. Z_2-even K-type projection

Compute: subset of Phase A K-types (Memorial Day enumeration ~30 nodes) that are Z_2-even under (x, θ) → (−x, θ+π)

Method: - Each K-type V_λ with weight (m, n) under K = SO(5)×SO(2) - Z_2-even condition: m + n even (or some specific parity rule) - Count Z_2-even K-types vs total - Compute gravitational source as sum over Z_2-even K-types

Connection to Lyra Lane E Dictionary: the Z_2-even K-types are likely the gravitationally-relevant SM particles (mass-carrying); Z_2-odd would be parity-violating (weak-sector).

Verification target: which K-types contribute to gravitational source, structurally?

7. Combined boundary integral

The full Hua-Korányi reproduction for gravitational source f_grav:

```
G_observed ~ ∫_{∂_S} P(z_obs, ζ) · f_grav(ζ) · dμ(ζ)
           = (S⁴ part) × (S¹ part) × (ℤ₂ projection of f_grav)
           = (4/3) · (S¹_correction) · (Z₂−even K−type sum) · G_2D_substrate
```

For the substantive numerical match:

```
G_observed / G_2D_substrate = (4/3) · (S¹_correction) · (Z₂−even projection)
6.674×10⁻¹¹ / 5.07×10⁻¹¹ = 1.316
(4/3) = 1.333

So: (S¹_correction) · (Z₂−even projection) = 1.316/1.333 = 0.987
                                             = 1 − 0.013   (~1.3% downward)
```

The full $S^1 \times \mathbb{Z}_2$ correction should be ~0.987, i.e., −1.3% from unity.

8. Candidate $S^1 \times \mathbb{Z}_2$ correction forms

Multiplicative factor ~0.987 in substrate primaries:

Form	Numerical	Match to 0.987
$1 - 1/\text{N\_max}$	0.993	close (1% too high)
$1 - 1/(\text{n\_C}\cdot\text{g})$	$1 - 1/35 = 0.971$	close (1.6% too low)
$1 - 1/\text{N\_max}^2 \times \text{N\_c}$	$1 - 3/18769 = 0.9998$	too small correction
$1 - 2/\text{N\_max}$	0.985	very close (0.2% off)
$1 - \pi^2/600$	$1 - 0.01645 = 0.984$	close (0.3% off)
$1 + \kappa\text{\_Bergman} \cdot \alpha^2$	$1 - 5/137^2 = 0.99973$	too small
$1 - \pi/\sqrt{(\text{N\_max} \cdot \text{N\_c}\cdot\text{g})}$	$1 - \pi/\sqrt{(2877)} = 0.942$	too low

Best candidates: $1 - 2/\text{N\_max}$ or $1 - \pi^2/600$ — both substrate-natural and ~0.985.

These are RECALLED candidates (Calibration #33). The actual derivation must come from the explicit S^1 boundary integral, not pattern-matching the residual.

9. Specific Elie Lane G-B targets

Refined from generic KK volume integral to specific Shilov boundary integrals:

G-B-1: Verify S^4 measure gives 4/3 via radial integration mechanism - Set up $\int$_{S^2} × $\int$_radial integration - Show (4/3) emerges from standard volume ratio

G-B-2: Compute $S^1/\mathbb{Z}_2$ Bergman-weighted phase integral - Evaluate $\int_0$^π K_factor(e^{iθ}) dθ explicitly using Bergman kernel formula - Get closed-form S^1_correction

G-B-3: Enumerate Z_2-even K-types in Phase A catalog - Apply (x,θ) → (−x,θ+π) parity to V_(m,n) K-types - List Z_2-even subset - Compute their substrate-source weight contribution

G-B-4: Combine and check - G_predicted = (4/3) × (G-B-2 result) × (G-B-3 weight) × G_2D_substrate - Compare to G_observed = 6.674×10^{-11} - Target: Tier 2 STRUCTURAL precision (0.01% to 1%)

10. Verification reading queue (Calibration #33 STANDING)

Pin specific theorems before promotion to VERIFIED-CITED:

1. Bergman kernel formula for D_IV^5 — Faraut-Korányi 1994 Ch. XII or Knapp 1986 Ch. VIII
2. Poisson-Szegő kernel coordinate form — FK Ch. XIII
3. Hua-Korányi reproduction formula — FK Ch. XIII
4. **Vol(∂_S D_IV^5) closed form** — FK Ch. XII
5. K-invariant measure normalization — FK Ch. XII or Hua 1963
6. Helgason κ_Bergman = -n_C for type IV — Helgason 1962 Ch. VIII Theorem 7.4 (or equivalent)

Keeper + Grace verification reading queue. Tier of citations: VERIFIED-CITED once read; current RECALLED.

11. Honest scope

What this framework provides: - Explicit structural decomposition: (4/3) $S^4 \times S^1$ correction × Z_2 projection - Concrete Elie Lane G-B computation targets (G-B-1 through G-B-4) - Substrate-mechanism reading: gravity emerges from Shilov boundary integration with specific factor structure - Verification path: Tier 2 STRUCTURAL precision check

What this framework does NOT provide: - Numerical evaluation of S^1 correction (multi-week Elie work) - Z_2-even K-type explicit enumeration (multi-week Lyra + Elie) - Verified citations (Calibration #33 reading pending) - Pattern-match shortcut closure (explicitly EXCLUDED)

12. Honest tier disposition

Framework: STRUCTURAL with explicit computation targets

Mechanism candidate: gravitational lift = (S^4 spatial) × (S^1 temporal correction) × (Z_2-even substrate source)

Numerical content: pending Elie G-B-1/2/3/4 multi-week computation

Promotion path: if G-B-4 lands at Tier 2 STRUCTURAL precision → K204 promotes from PARTIAL to STRUCTURAL → G derived from substrate at structurally-honest tier

Cal cold-read priority: Cal #192 candidate when framework absorbed by Lyra/Elie.

13. Cross-references

- `Keeper_G_Substrate_Clean_4_Step_Derivation_Framework.md` — overall 4-step chain (this refines Step B)
- `Keeper_K204_partial_kappa_Bergman_n_C_Audit.md` — κ_Bergman = -n_C foundation
- `Keeper_K200_Tier0_v0_1_6_AuditFramework.md` — gates G1-G5 (this advances G5 specifically)
- `Keeper_Sphere_Reconciliation_Analysis_G2.md` — sphere geometry (this further refines)
- T2442 c_FK = 225/π^{9/2} RATIFIED (Saturday)
- Faraut-Korányi 1994 Ch. XII-XIII (verification queue)

14. The honest creative reframe

Casey's insight reframes the G chain:

Before: generic KK reduction G_4 = G_substrate / V_6 (Step A vague)

Now: explicit Shilov boundary integration with: - S^4 geometric lift (4/3 disk-to-ball + radial integration) - S^1 phase correction (Bergman-weighted phase integral) - $\mathbb{Z}_2$-even K-type projection (gravitational K-type selection)

This is the substantive creative path the 24% gap opened. The factor 4/3 has geometric mechanism (disk-to-ball lift, not arithmetic identity); the few-percent residual has structural location (S^1 boundary integration); the substrate-source has Z_2-even constraint (specific K-type subset).

Multi-week Elie + Lyra joint work closes G chain at honest Tier 2 STRUCTURAL if successful.

— Keeper. Boundary integral investigation framework filed. Standard math RECALLED with explicit verification reading queue (Calibration #33). Casey's S^4/S^1 factorization insight captured at structural level. Elie Lane G-B refined to specific Shilov boundary computations G-B-1/2/3/4. Standing for team absorption + multi-week computation work.